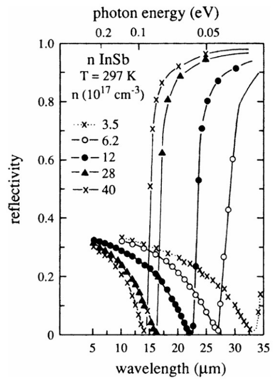
CHAPTER 12

Optical Properties of Plasmons,

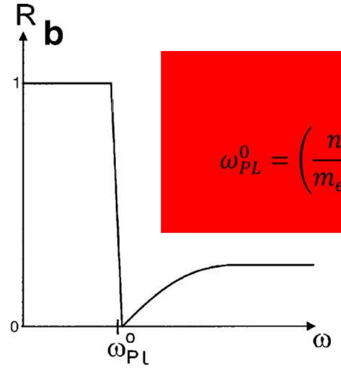
Plasmon-Phonon Mixed States and of Magnons

一定要会

- Plasmons as collective excitations of the carriers in a partly filled band usually do not exist in intrinsic semiconductors under thermodynamic equilibrium
- Two conditions under which plasmons can be observed, high doping or high excitation
  - In highly doped semiconductors, there is a high density either of electrons or of holes
  - In highly excited semiconductors, an electron–hole plasma can be formed, which consists of electrons and holes



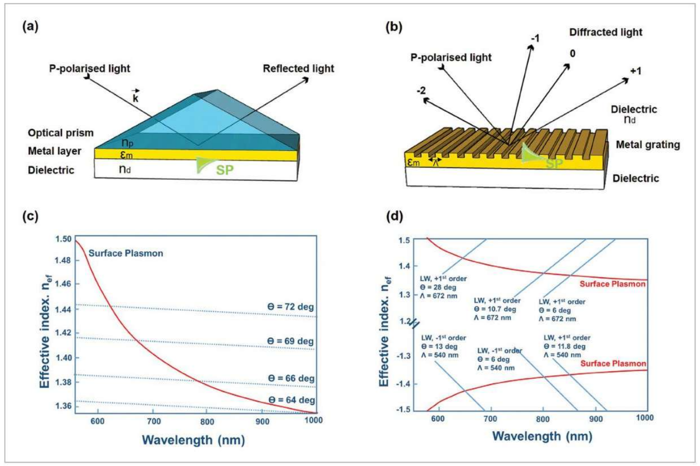
The reflection spectrum in the vicinity of the plasmon resonance in doped InSb samples



$$\omega_{PL}^{0} = \left(\frac{ne^2}{m_e \varepsilon \varepsilon_0}\right)^{\frac{1}{2}}$$

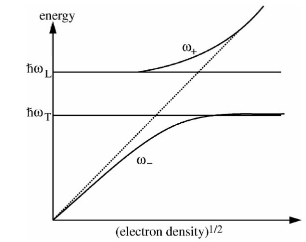
12.1 Surface Plasmons

- Surface plasmons are electron collective oscillations that exist at the interface between a metal and a dielectric material.
- The surface plasmons can be investigated either by attenuated total reflection or by a grating with spacing Λ engraved in the surface.



12.2 Plasmon-Phonon Mixed States

- When $\omega_{PL} \approx \omega_{LO}$ there are plasmon-phonon mixed states



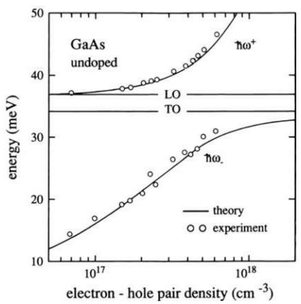


Fig. 12.3. The dependence of the plasmon-phonon mixed mode on the electron–hole pair density in highly photon-excited GaAs [84N1]

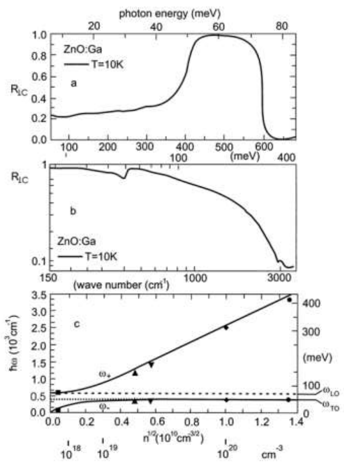


Fig. 12.4. Reflection spectra of n-doped ZnO:Ga samples with two different doping levels (note the different abscissas) (**a**,**b**) and the various transverse and longitudinal eigenenergeis of the plasmon-phonon mixed states as a function of carrier density (c) [97G1]

12.3 Plasmons in Systems of Reduced Dimensionality

- In Fig. 12.5 we show two different modulation-doped GaAs/$Al_{1-y}Ga_yAs$ samples.
- The data points have been deduced from angle-resolved Raman scattering, which allows variation of k_P
- If the quantum wells couple in a MQW, also a transverse wavelength or $\boldsymbol{k}_\perp$
- A dispersion for $\omega_{PL}(k_P, k_\perp)$ taking such effects into account is given:

$$\omega_{PL} = \left\{ \frac{2\pi n e^2}{\varepsilon_M m^*_{eff}} \boldsymbol{k}_P \frac{\sin k_P d}{\cosh k_P d - \cos k_\perp d} \right\}^{1/2} \quad (12.4)$$

- The solid lines have been calculated with (12.4) and the dashed lines are linear approximations.

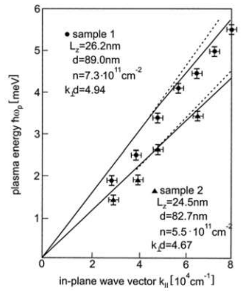


Fig. 12.5. The in-plane dispersion relation of plasmons in two different modulation doped GaAs/$Al_{1-y}Ga_yAs$ samples [82O1]